

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Knowledge Representation Design
Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning. (BTL 3)*

Assessment Tool Weightage: 40 %

Total Marks: 100

Time: 60 Mins

No. of Questions: 5

Annexure A
Knowledge Representation Design Assessment

Code of Conduct:

I T Bhupal/192425243 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:*bhupalreddy*

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Knowledge Engineering and Knowledge Base Design	15%	
3. Propositional Logic Representation	10%	
4. First-Order Logic Representation	15%	
5. Unification and Resolution	15%	
6. Forward and Backward Chaining	15%	
7. Inference and Reasoning Accuracy	10%	
8. System Architecture / Representation Diagram	5%	
9. Prototype Implementation and Results	5%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE QUESTION

1. Smart Healthcare Diagnosis System

A hospital wants to develop an AI-based system that can identify possible illnesses from patient symptoms. The system contains facts such as:

- Ravi has fever.
- Ravi has cough.
- Ravi has body pain.
- Patients with fever and cough may have flu.
- Patients with flu and body pain may have viral infection.
- Patients with viral infection should be advised to rest.

Task:

1. Represent the given knowledge using **Propositional Logic**.
2. Convert the knowledge into **First-Order Logic (FOL)**.
3. Construct a knowledge base containing facts and rules.
4. Demonstrate **unification** for at least two predicates.
5. Use **forward chaining** to determine Ravi's condition.
6. Use **backward chaining** to prove whether Ravi needs rest.
7. Demonstrate **resolution** for one selected query.
8. Explain the final inference produced by the system.

2. Intelligent Loan Approval System

A bank wants to develop an AI system to determine whether a customer is eligible for a loan. The knowledge base contains information about income, credit score, employment, existing loans, and repayment history.

For example:

- Arun has a stable job.
- Arun has a high credit score.
- Arun has sufficient income.
- Customers with stable employment and sufficient income are eligible for preliminary approval.
- Customers with high credit scores and good repayment history are considered low-risk.
- Low-risk customers with preliminary approval can be recommended for loan approval.

Task:

1. Identify the **entities, facts, predicates, and rules** involved.
2. Represent the knowledge using **First-Order Logic**.
3. Build a knowledge base with at least **10 facts and 8 rules**.
4. Demonstrate **unification and substitution**.

5. Apply **forward chaining** to determine whether Arun can be recommended for approval.
6. Apply **backward chaining** for the query `LoanApproved(Arun)`.
7. Use **resolution** to prove or disprove one query.
8. Discuss how the knowledge-engineering process helps maintain the loan rules.

3. Intelligent Vehicle Fault Diagnosis

An automobile service center wants to develop an expert system that identifies possible vehicle faults based on symptoms.

The system knows that:

- A vehicle has a dead battery if it does not start and the headlights are dim.
- A vehicle may have a fuel problem if the engine cranks but does not start.
- A vehicle may have an overheating problem if the temperature is high and coolant is low.
- A vehicle should be sent for immediate service if overheating is detected.

For a vehicle named **Car101**, the system receives:

- Car101 does not start.
- Car101 has dim headlights.
- Car101 has a high temperature.

Task:

1. Design the **knowledge representation** for the vehicle diagnosis system.
2. Represent at least five rules using **FOL**.
3. Represent selected statements using **Propositional Logic**.
4. Demonstrate **unification** using suitable predicates.
5. Apply **forward chaining** to identify possible faults.
6. Apply **backward chaining** for the query `NeedsService(Car101)`.
7. Demonstrate **resolution** for one diagnosis.
8. Explain how inference is used to arrive at the final diagnosis.

4. Intelligent Student Academic Advising System

A university wants an AI-based academic advising system that recommends actions to students based on their academic performance.

The system contains rules such as:

- Students with attendance below 75% require attendance counseling.
- Students who fail two or more courses require academic counseling.
- Students with high attendance and good academic performance can be recommended for advanced courses.
- Students who have completed prerequisite courses can register for the corresponding advanced course.

Consider a student **Priya** who has:

- Attendance of 68%.
- Two failed courses.
- Completed the prerequisite for an advanced AI course.

Task:

1. Identify the **knowledge engineering requirements** for the system.
2. Represent the academic rules using **FOL**.
3. Create a knowledge base containing facts and rules.
4. Demonstrate **unification** using student and course predicates.
5. Use **forward chaining** to identify recommendations for Priya.
6. Use **backward chaining** to determine whether Priya requires academic counseling.
7. Apply **resolution** to prove one selected recommendation.
8. Explain the advantages and limitations of using a rule-based inference system for academic advising.

5. Cybersecurity Threat Detection System

An organization wants to develop an AI knowledge-based system to identify suspicious activities in its network.

The knowledge base contains rules such as:

- Multiple failed login attempts from the same user may indicate a brute-force attack.
- A login from an unusual location combined with multiple failed attempts indicates suspicious activity.
- Suspicious activity followed by unauthorized file access indicates a possible security breach.
- A possible security breach should generate a high-priority alert.

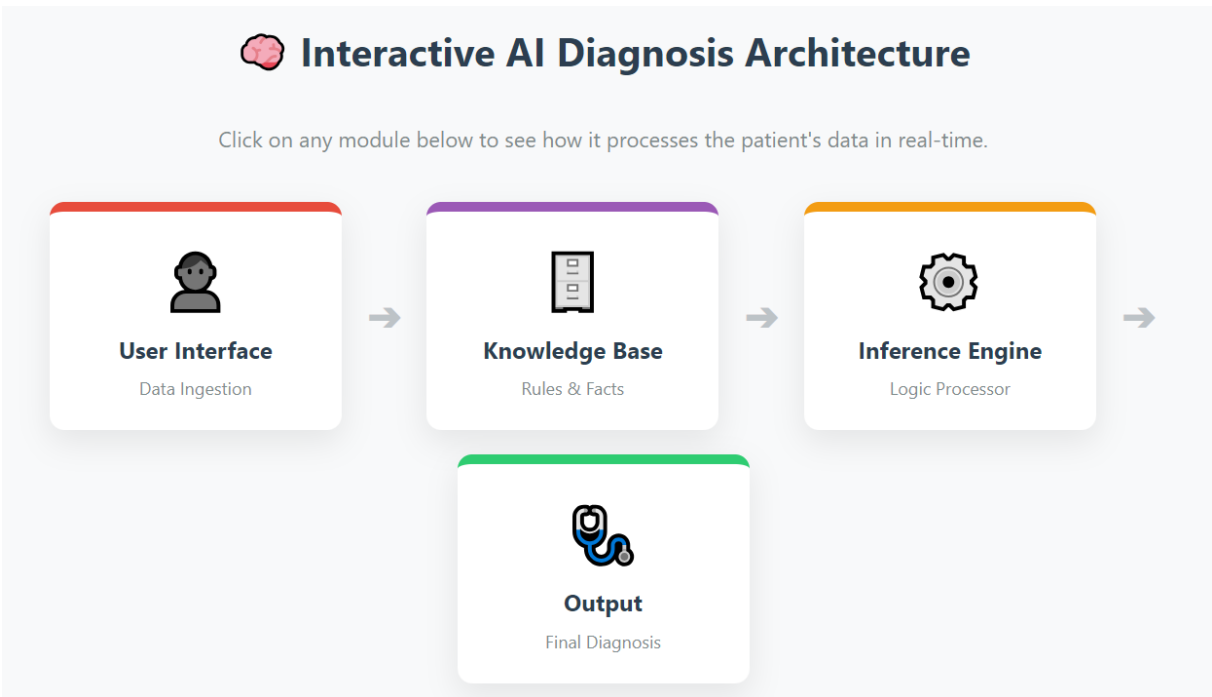
For user **User101**, the system observes:

- Five failed login attempts.
- Login from an unusual location.
- Unauthorized access to a confidential file.

Task:

1. Design a **knowledge representation model** for the cybersecurity domain.
2. Represent the knowledge using **Propositional Logic and FOL**.
3. Construct at least **10 facts and 8 rules**.
4. Demonstrate **unification and substitution**.
5. Apply **forward chaining** to determine whether a security alert should be generated.
6. Apply **backward chaining** for the query `HighPriorityAlert(User101)`.
7. Use **resolution** to prove the existence of a possible security breach.
8. Compare **forward chaining and backward chaining** for cybersecurity reasoning and justify which is more suitable.

Smart Healthcare Diagnosis System: AI Reasoning Report



1. Problem Analysis and Understanding

Objective: To develop an AI-driven healthcare system capable of diagnosing illnesses and advising treatments based on a set of observed symptoms.

- **Entities:** Ravi (Patient)
- **Symptoms (Facts):** Fever, Cough, Body Pain
- **Conditions (Inferences):** Flu, Viral Infection
- **Action/Recommendation:** Advise Rest
- **Logical Flow:** The system must map initial symptoms to an intermediate condition (Flu), combine that with another symptom (Body Pain) to find a final diagnosis (Viral Infection), and map the diagnosis to a prescription (Rest).

 **User Interface (Data Ingestion)**

Purpose: Captures raw observations about the patient and converts them into logical propositions.

Current Inputs translated to Facts:

- `Has(Ravi, Fever)`
- `Has(Ravi, Cough)`
- `Has(Ravi, BodyPain)`

These facts are immediately sent to the Knowledge Base to initialize the Working Memory.

2. Task 1: Propositional Logic (PL) Representation

In Propositional Logic, we use simple, declarative statements (propositions) without variables.

Propositional Variables for Ravi:

- F = Ravi has a fever.
- C = Ravi has a cough.
- B = Ravi has body pain.
- Fl = Ravi has the flu.
- V = Ravi has a viral infection.
- R = Ravi is advised to rest.

Knowledge Represented in PL:

- **Facts:** F, C, B
- **Rules:**
 - $F \wedge C \rightarrow Fl$
 - $Fl \wedge B \rightarrow V$
 - $V \rightarrow R$

3. Task 2: First-Order Logic (FOL) Representation

First-Order Logic (FOL) allows us to generalize the rules for any patient using variables, constants, predicates, and quantifiers.

- **Constant:** Ravi
- **Variables:** x (representing any patient)
- **Predicates:** $Fever(x), Cough(x), BodyPain(x), Flu(x), ViralInfection(x), NeedsRest(x)$

Knowledge Represented in FOL:

- **Facts:**
 - $Fever(Ravi)$
 - $Cough(Ravi)$
 - $BodyPain(Ravi)$
- **Rules:**
 - $\forall x (Fever(x) \wedge Cough(x) \rightarrow Flu(x))$
 - $\forall x (Flu(x) \wedge BodyPain(x) \rightarrow ViralInfection(x))$
 - $\forall x (ViralInfection(x) \rightarrow NeedsRest(x))$

4. Task 3: Knowledge Base (KB) Construction

The Knowledge Base acts as the core memory for the inference engine, separated into Facts (Axioms) and Rules (Implications).

Knowledge Base (Ontology & Rules)

Purpose: Stores the medical domain logic using First-Order Logic (FOL).

Rule Base:

- **Rule 1:** $\forall x (Has(x, Fever) \wedge Has(x, Cough) \rightarrow Disease(x, Flu))$
- **Rule 2:** $\forall x (Disease(x, Flu) \wedge Has(x, BodyPain) \rightarrow Disease(x, ViralInfection))$
- **Rule 3:** $\forall x (Disease(x, ViralInfection) \rightarrow Advice(x, Rest))$

Facts in KB:

- **Fact 1:** Fever(Ravi)
- **Fact 2:** Cough(Ravi)
- **Fact 3:** BodyPain(Ravi)

Rules in KB:

- **Rule 1:** $\forall x (Fever(x) \wedge Cough(x) \rightarrow Flu(x))$
- **Rule 2:** $\forall x (Flu(x) \wedge BodyPain(x) \rightarrow ViralInfection(x))$
- **Rule 3:** $\forall x (ViralInfection(x) \rightarrow NeedsRest(x))$

5. Task 4: Unification

Unification is the process of making two different logical expressions identical by finding a substitution for their variables. We want to apply Rule 1 to our known facts to infer new information.

- **Predicate from Rule 1 Premise:** Fever(x)
- **Predicate from KB Fact 1:** Fever(Ravi)
- **Most General Unifier (MGU):** $\theta = \{x / Ravi\}$

Applying this substitution to the entire Rule 1 yields: Fever(Ravi) $\wedge$ Cough(Ravi) $\rightarrow$ Flu(Ravi)

6. Task 5: Forward Chaining

Forward chaining is a data-driven approach, starting with known facts and applying rules to extract new facts until a final condition is met.

Inference Engine (Reasoning Core)

Purpose: Applies logic algorithms to derive new information.

Executing Forward Chaining:

1. **Unification:** The system substitutes the variable `x` with the constant `Ravi`.
2. **Iteration 1:** Matches `Fever` and `Cough` with Rule 1 → Derives `Flu`.
3. **Iteration 2:** Matches `Flu` and `BodyPain` with Rule 2 → Derives `Viral Infection`.
4. **Iteration 3:** Matches `Viral Infection` with Rule 3 → Derives `Advise Rest`.

Execution Steps:

- **Iteration 1:** We know `Fever(Ravi)` and `Cough(Ravi)`. Applying Rule 1 with MGU $\{x / \text{Ravi}\}$ derives the new inferred fact: **`Flu(Ravi)`**.
- **Iteration 2:** We know `Flu(Ravi)` (newly inferred) and `BodyPain(Ravi)` (Fact 3). Applying Rule 2 with MGU $\{x / \text{Ravi}\}$ derives the new inferred fact: **`ViralInfection(Ravi)`**.
- **Iteration 3:** We know `ViralInfection(Ravi)` (newly inferred). Applying Rule 3 with MGU $\{x / \text{Ravi}\}$ derives the new inferred fact: **`NeedsRest(Ravi)`**.

Conclusion: Ravi's condition is a Viral Infection.

7. Task 6: Backward Chaining

Backward chaining is a goal-driven approach. We start with the goal and work backward to see if the facts in the KB support it.

Goal: Prove `NeedsRest(Ravi)`

Execution Steps:

- **Step 1:** Does `NeedsRest(Ravi)` exist in KB? No.
- **Step 2:** Find a rule concluding `NeedsRest(x)`. Rule 3 concludes `NeedsRest(x)` if `ViralInfection(x)`. **New Sub-goal:** Prove `ViralInfection(Ravi)` (using substitution $\{x / \text{Ravi}\}$).
- **Step 3:** Does `ViralInfection(Ravi)` exist in KB? No.
- **Step 4:** Find a rule concluding `ViralInfection(x)`. Rule 2 concludes `ViralInfection(x)` if `Flu(x) ∧ BodyPain(x)`. **New Sub-goals:** Prove `Flu(Ravi)` AND prove `BodyPain(Ravi)`.
- **Step 5:** Check `BodyPain(Ravi)`: It exists in KB (Fact 3). Sub-goal met.
- **Step 6:** Check `Flu(Ravi)`: Does not exist in KB. Find a rule concluding `Flu(x)`. Rule 1 concludes `Flu(x)` if `Fever(x) ∧ Cough(x)`. **New Sub-goals:** Prove `Fever(Ravi)` AND prove `Cough(Ravi)`.
- **Step 7:** Check `Fever(Ravi)`: It exists in KB (Fact 1). Sub-goal met.

- **Step 8:** Check Cough(Ravi): It exists in KB (Fact 2). Sub-goal met.

Since all fundamental sub-goals bottom out into known facts, the original goal NeedsRest(Ravi) is **successfully proven**.

8. Task 7: Resolution

Resolution requires converting the KB into Conjunctive Normal Form (CNF) and negating the goal to find a contradiction (Proof by Refutation).

Step 1: Convert KB to CNF (Eliminating implications $P \rightarrow Q \equiv \neg P \vee Q$)

1. Fever(Ravi)
2. Cough(Ravi)
3. BodyPain(Ravi)
4. $\neg\text{Fever}(x) \vee \neg\text{Cough}(x) \vee \text{Flu}(x)$
5. $\neg\text{Flu}(y) \vee \neg\text{BodyPain}(y) \vee \text{ViralInfection}(y)$
6. $\neg\text{ViralInfection}(z) \vee \text{NeedsRest}(z)$

Step 2: Negate the Goal 7. $\neg\text{NeedsRest}(\text{Ravi})$

Step 3: Resolve systematically to find a contradiction

- **Resolve (6) and (7)** with MGU $\{z / \text{Ravi}\}$: Derives $\neg\text{ViralInfection}(\text{Ravi})$ (8)
- **Resolve (5) and (8)** with MGU $\{y / \text{Ravi}\}$: Derives $\neg\text{Flu}(\text{Ravi}) \vee \neg\text{BodyPain}(\text{Ravi})$ (9)
- **Resolve (9) and (3)**: Derives $\neg\text{Flu}(\text{Ravi})$ (10)
- **Resolve (4) and (10)** with MGU $\{x / \text{Ravi}\}$: Derives $\neg\text{Fever}(\text{Ravi}) \vee \neg\text{Cough}(\text{Ravi})$ (11)
- **Resolve (11) and (1)**: Derives $\neg\text{Cough}(\text{Ravi})$ (12)
- **Resolve (12) and (2)**: Derives $\perp$ (Empty Clause / Contradiction)

Because we reached a contradiction, the negated goal is false, which proves the original goal NeedsRest(Ravi) is **True**.

9. Task 8: Final Inference

Final Output (Actionable Inference)

Conclusion Validated: Ravi is diagnosed with a Viral Infection.

System Recommendation: Ravi is advised to Rest.

The system successfully completed its reasoning cycle. By using backward chaining (Resolution), we can also logically prove this output contains zero contradictions.

Based on the systemic processing of logical rules via Forward Chaining, Backward Chaining, and Refutation Resolution, the Smart Healthcare AI System successfully identifies that:

- The combination of Ravi's fever and cough indicates he has the **flu**.
- The presence of the flu alongside his body pain escalates the diagnosis to a **viral infection**.
- Because patients with viral infections must rest, the ultimate medical inference produced by the system is that **Ravi must be advised to rest to recover from a viral infection**.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
Problem Analysis and Understanding	15%	13–15% – Clearly understands and analyzes the real-world problem, objectives, entities, facts, constraints, and expected inference.	10–12% – Understands the problem with minor omissions.	7–9% – Shows partial understanding with some important elements missing.	0–6% – Poor or incorrect understanding of the problem.
Knowledge Engineering & Knowledge Base Design	15%	13–15% – Develops a complete, consistent, and well-structured knowledge base with appropriate facts, predicates, entities, and rules.	10–12% – Knowledge base is well designed with minor inconsistencies or omissions.	7–9% – Basic knowledge base is developed but several facts/rules are missing or unclear.	0–6% – Knowledge base is incomplete, inconsistent, or inappropriate.
Propositional & First-Order Logic Representation	20%	17–20% – Correctly represents the domain using propositions, predicates, variables, constants, quantifiers, and logical connectives with excellent accuracy.	13–16% – Logic representation is mostly correct with minor errors.	9–12% – Provides basic representations but has several logical or syntactic errors.	0–8% – Logic representation is incorrect, incomplete, or missing.
Unification & Resolution	15%	13–15% – Correctly performs substitutions, identifies the MGU, and applies resolution systematically to derive valid conclusions.	10–12% – Applies unification and resolution correctly with minor errors.	7–9% – Demonstrates partial understanding but misses important steps or contains errors.	0–6% – Unable to correctly apply unification or resolution.

Forward Chaining & Backward Chaining	15%	13–15% – Correctly applies both forward and backward chaining with clear step-by-step reasoning and appropriate rule selection.	10–12% – Correctly applies both methods with minor errors or omissions.	7–9% – Demonstrates partial understanding of one or both chaining methods.	0–6% – Incorrectly applies or fails to demonstrate chaining techniques.
Inference & Reasoning Accuracy	10%	9–10% – Produces logically valid conclusions and clearly explains how facts and rules lead to the final inference.	7–8% – Conclusions are mostly correct with minor reasoning gaps.	5–6% – Some correct inferences but reasoning is incomplete or unclear.	0–4% – Conclusions are incorrect, unsupported, or missing.
Knowledge Representation Diagram / System Architecture	5%	5% – Provides a complete, accurate, well-labelled diagram showing knowledge base, inference engine, reasoning process, and output.	4% – Diagram is mostly correct with minor omissions.	3% – Diagram is partially complete or lacks clarity.	0–2% – Diagram is missing or incorrect.
Originality, Critical Thinking & Presentation	5%	5% – Demonstrates excellent independent thinking, appropriate real-world application, comparison of reasoning approaches, and clear presentation.	4% – Demonstrates good reasoning and clear presentation.	3% – Shows limited critical thinking and basic presentation.	0–2% – Shows little originality, weak reasoning, or poor presentation.